

Bhalaji Yadav Kantepalle

Richmond, VA | +1 (804) 310-4169 | kbhalajiyadav22@gmail.com | linkedin.com/in/kbhalajiyadav

PROFESSIONAL SUMMARY

Chemical Engineer with **1.5+ years of experience** bridging **Materials Science** and **Regulated Quality Operations**. Proven ability to accelerate medical product innovation by translating complex R&D data into commercial value, evidenced by successful **grant funding** for wearable diagnostics and **US FDA audit readiness**. Expert in **Material Characterization** (Adhesion, Fatigue) and Process Validation (V-Model, CAPA). Passionate about advancing the world of health through reliable, patient-centric material solutions.

EDUCATION

Virginia Commonwealth University

M.S., Chemical & Life Science Engineering | **GPA: 3.59**

Richmond, VA
Aug 2024 – May 2026

Achievements: Graduate Assistantship Award (Fall 2025) – **Full Tuition Waiver & Stipend**

Amrita Vishwa Vidyapeetham (Deemed University)

B.Tech., Chemical Engineering | **CGPA: 2.75/4.0**

Coimbatore, Tamil Nadu
Jul 2018 – Jun 2022

Narayana Junior College

Higher Secondary Diploma | **Score: 803/1000**

Tirupati, Andhra Pradesh
Jun 2016 – Mar 2018

Sri Chaitanya EM High School

High School Diploma | **GPA: 7.8/10**

Madanapalle, Andhra Pradesh
May 2015 – Apr 2016

TECHNICAL SKILLS

Research & Strategy: Technical Writing, Literature Review, Grant Writing, Market Intelligence, FDA Regulatory Analysis (DMF)

Pharma Compliance: cGMP Guidelines, CAPA Management, Root Cause Analysis (RCA), Audit Preparation (US FDA)

Material Science: Polymer Physics, Rheology, Liquid Crystals, Adhesion Testing (ASTM D2724), Microscopy, NMR, UV-Vis

Data Tools: Python (Data Mining), SAP Business One (ERP), Power Automate, Tableau, Docker

RESEARCH EXPERIENCE

VCU Soft Functional Materials Lab

Graduate Research Assistant

Richmond, VA
Sep 2024 – Present

- Drafted a successful \$30,000 in Grant proposal** (Commonwealth Cyber Initiative) by conducting a comprehensive **literature review** to substantiate the clinical viability of thermochromic textiles for **diabetic inflammation monitoring**.
- Engineered an Integrated Optical Metrology System** (Python/OpenCV) that synchronizes mechanical strain with colorimetry, demonstrating strategic thinking to eliminate temporal artifacts and **optimize 35+ metrics** for thermodynamic reversibility (increasing throughput **>300%**).
- Predicted Device Fatigue Limits** by decoupling elastic signals from plastic damage using **unsupervised Machine Learning (PCA)**, identifying failure modes to enhance material longevity in dynamic applications.
- Validated structural changes** by extending the **Michel-Lévy interference chart**, utilizing **rheology models** (Cross/Power-law) to correlate optical data with physical material properties.
- Engineered a high-throughput Fracture Signal Analysis protocol** for adhesion testing that automates **polarity rectification** and **ASTM-compliant plateau detection**, reducing analysis time by **>80%** while eliminating manual bias.

INDUSTRY EXPERIENCE

Kreative Organics Pvt. Ltd.

Technical Project Manager

Hyderabad, India
May 2023 – May 2024

- Rapid Career Progression:** Promoted from Intern to Manager in 6 months for demonstrating high **adaptability** in navigating pharmaceutical regulations and operationalizing Quality Management Systems.
- Directed a **CAPA/Root Cause Analysis** initiative to re-engineer validation logic, defining a V-Model roadmap that ensured full readiness for **US FDA audits** and compliance with cGMP standards.
- Synthesized supply chain intelligence to uncover high-value market gaps, conducting deep analysis of **FDA Drug Master Files (DMF)** to steer strategic **New Market Entry** for regulated materials.
- Spearheaded SAP System Requalification**, executing a precision **5-hour overnight cutover** to safeguard production continuity and data integrity, minimizing operational risk.

Technical Project Intern

Oct 2022 – Apr 2023

- Engineered a **Supply Chain Optimization Workflow** that cross-referenced global trade data with **competitor benchmarks**, cutting research time by **70%** to identify viable sales targets.
- Designed and Containerized** a Standardized Chemical Data Protocol (Python/Docker) to eliminate naming inconsistencies, driving a **90% uplift** in data retrieval accuracy for technical teams.
- Designed a **Stage-Gate aligned** Project Management Framework (Notion) to coordinate cross-functional handoffs, which was subsequently **adopted company-wide** to streamline R&D timelines.

PUBLICATIONS & ACHIEVEMENTS

Journal Article: I. Caloian, J. Trapp, **B. Y. Kantepalle**, et al. "Mechanical Properties of Dual-Layer Electrospun Fiber Mats". *Polymers* 17(13), 1777 (2025). [doi:10.3390/polym17131777]

Selected Participant: 1st National Neutron Scattering School, Oak Ridge National Laboratory (Sep 2025).

Poster Presentation: "Adhesives for Personalized Wearable Devices" (ACS Fall 2025, VCU Research Symposium - Apr 2025, VCU Engineering Research Showcase - Mar 2025). B. Y. Kantepalle, H. Stwodah, D. Joung, C. Tang.